

Chapter 27

Presuppositions

(Joint work with Koji Mineshima)

A sentence's meaning is often analyzed as comprising its foreground and background components. The cleft sentence in (714) serves as a classic illustration of this distinction.

(714) It is John who broke my iPhone.

The background (or non-at-issue) content of (714) is *someone broke my iPhone*, which the speaker assumes is mutually accepted as true by the hearer. The foreground (or at-issue) content of (714) is *John was the one who did it*, which conveys the primary communicative intent.

The background contents of a sentence are known to pose a challenge in semantic theory. Unlike foreground contents, a certain kind of background content is inferred from both the positive and negative forms of the original sentence, as demonstrated in (715a) and (715b).

- (715) a. It is John who broke my iPhone. $\implies$ Someone broke my iPhone.
 b. It is not John who broke my iPhone. $\implies$ Someone broke my iPhone.

The challenge here lies in the fact that the inference relation $\implies$ presented above cannot be analyzed as *entailment*: for it were, the two inferences would take the following form.

(716) $A \vdash B, \neg A \vdash B$

In a classical setting where the law of the excluded middle ($\vdash A + \neg A$) holds, (716) would lead to an absurd conclusion $\vdash B$, namely, B is true under any context. This means that *someone broke my iPhone* is a logical theorem, which is clearly not the case. Therefore, the inference relation $\implies$ observed in (715a) and (715b) must be a distinct mode of inference from classical entailment. Semanticists termed this mode as *presupposition*, and accordingly describe the relation in (715a) and (715b) as *It is John who broke my iPhone* and *It is not John who broke my iPhone* presuppose *Someone broke my iPhone*. Moreover, when discussing the inference in (715b), we often describe the background content as *projecting out of*, or *surviving in* the negation.

27.1 Presupposition Projection

The background contents not only project out of negation, but also out of other embedded contexts: (717) is inferred from every sentence in (718).

- (717) Someone broke my iPhone.
- (718)
- a. It wasn't John who broke my iPhone. negation
 - b. Maybe it was John who broke my iPhone. modal
 - c. If it was John who broke my iPhone, then he has to pay for it.
the antecedent of a conditional
 - d. Was it John who broke my iPhone? question
 - e. Suppose that it was John who broke my iPhone.
hypothetical assumption

These examples stand in stark contrast to the case of entailment as exemplified below.

- (719) John is an American pianist. $\implies$ John is American.

This entailment does not survive in the environments (720a-e), in contrast to the case of (718a-e).

- (720)
- a. John is not an American pianist. negation
 - b. Maybe John is an American pianist. modal
 - c. If John is an American pianist, he is skillful.
the antecedent of a conditional
 - d. Is John an American pianist? question
 - e. Suppose that John is an American pianist.
hypothetical assumption

Standard semantics in modal FoL correctly predicts the inference patterns in (720a-c).

- (719) John is an American pianist.
american(john) $\wedge$ pianist(john)
- (720)
- a. John is not an American pianist.
 $\neg(\mathbf{american(john) \wedge pianist(john)})$
 - b. Maybe John is an American pianist.
 $\Diamond(\mathbf{american(john) \wedge pianist(john)})$
 - c. If John is an American pianist, he is skillful.
american(john) $\wedge$ pianist(john) $\rightarrow$ skillful(john)

Therefore, we obtain the following.

- (721)
- a. (719) $\vdash$ **american(john)**
 - b. (720a) $\not\vdash$ **american(john)**

- c. (720*b*) $\not\models$ **american**(john)
- d. (720*c*) $\not\models$ **american**(john)

The presupposition, however, shows a different behaviour.

- (714) It was John who broke my iPhone. SR_1
- (718) a. It wasn't John who broke my iPhone. $\neg SR_1$
- b. Maybe it was John who broke my iPhone. $\Diamond SR_1$
- c. If it was John who broke my iPhone, he has to fix it. $SR_1 \rightarrow \dots$

Now the question is what SR accounts for the following inference patterns.

- (722) a. $SR_1 \models \exists x(\mathbf{broke}(x, \mathbf{myiPhone}))$
- b. $\neg SR_1 \models \exists x(\mathbf{broke}(x, \mathbf{myiPhone}))$
- c. $\Diamond SR_1 \models \exists x(\mathbf{broke}(x, \mathbf{myiPhone}))$
- d. $SR_1 \rightarrow A \models \exists x(\mathbf{broke}(x, \mathbf{myiPhone}))$

27.2 Presupposition triggers

Beyond the cleft construction, other constructions also *triggers* presuppositions. Below is a comprehensive, though not exhaustive, list of the presupposition triggers.*¹

- | | | |
|-------|--|---------------------|
| (723) | a. The elevator in this building is clean.
b. There is an elevator in this building. | Description |
| (724) | a. John's sister is happy.
b. John has a sister. | Possessive |
| (725) | a. Bill regrets that he lied to Mary.
b. Bill lied to Mary. | Factive |
| (726) | a. John has stopped beating his wife.
b. John has beaten his wife. | Aspectual |
| (727) | a. Harry managed to find the book.
b. Finding the book required some effort. | Implicative |
| (728) | a. Sam broke the window again today.
b. Sam broke the window before. | Iterative |
| (729) | a. It was Sam who broke the window.
b. Someone broke the window. | Cleft |
| (730) | a. What John broke was his typewriter.
b. John broke something. | Pseudo-cleft |
| (731) | a. $[\text{Pat}]_F$ is leaving, too . (Focus on <i>Pat</i>)
b. Someone other than Pat is leaving. | Additive |

*¹ For classical examples of presupposition triggers, see Levinson (1983), Soames (1989), Geurts (1999), and Beaver (2001), among others.

27.3 DTS on Presupposition

The projection of presupposition is naturally accounted for using DTS, using underspecified types for representing the lexical meaning of presupposition triggers. As a preparation, let us assume that the lexical entry for the determiner *the* is specified as follows.

$$(732) \quad \llbracket \text{the} \vdash NP^*/N \rrbracket \stackrel{\text{def}}{=} \lambda n. \left[\begin{array}{c} u@ \left[\begin{array}{c} x : e \\ n(x) \end{array} \right] \\ \boxed{\pi_1 u} \end{array} \right]$$

The syntactic structure and semantic composition of

(733) The king of France is wise.

are given as follows.

(734) Syntactic structure of (733)

$$\frac{\frac{\frac{\text{the}}{NP^*/N} \quad \frac{\frac{\text{king}}{N/PP_{of}} \quad \frac{\frac{\text{of}}{PP_{of}/NP} \quad \frac{\text{France}}{NP}}{PP_{of}}}{N}}{NP^*} > \quad \frac{\frac{\text{is}}{(S \setminus NP / (S \setminus NP))^*} \quad \frac{\text{wise}}{(S \setminus NP)^*}}{(S \setminus NP)^*} <^*}{\frac{S^*}{S} \checkmark} >^*$$

(735) Semantic composition of (733)

$$\frac{\lambda n. \left[\begin{array}{c} \text{the} \\ u@ \left[\begin{array}{c} x : e \\ n(x) \end{array} \right] \\ \boxed{\pi_1 u} \end{array} \right] \quad \frac{\frac{\text{king}}{\lambda z. \lambda x. \mathbf{kingOf}(x, z)} \quad \frac{\frac{\text{of}}{id} \quad \frac{\text{France}}{fr}}{fr} >}{\lambda x. \mathbf{kingOf}(x, fr)} > \quad \frac{\frac{\text{is}}{id} \quad \frac{\text{wise}}{\lambda x. \mathbf{wise}_s(x)}}{\lambda x. \mathbf{wise}_s(x)} >}{\left[\begin{array}{c} u@ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \\ \boxed{\pi_1 u} \end{array} \right] \quad \frac{\left[\begin{array}{c} u@ \left[\begin{array}{c} y : e \\ \mathbf{KingOf}(y, fr) \end{array} \right] \\ \boxed{\mathbf{wise}_s(\pi_1 u)} \end{array} \right]}{\left[\begin{array}{c} u@ \left[\begin{array}{c} y : e \\ \mathbf{KingOf}(y, fr) \end{array} \right] \\ \mathbf{wise}_s(\pi_1 u) \end{array} \right]} \checkmark} <^* \checkmark$$

How, then, might we account for the inference of the following “background content” from (733)?

(736) There exists a French king.

For the sake of simplicity, let us proceed by assuming that the semantic representation of (736) is a type that is provably equivalent to the following.

$$(737) \quad \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right]$$

The well-formedness of the sentence in (733) is verifiable by the following type-check.

(738) Type check diagram of (733)

$$\frac{\begin{array}{c} \vdots \\ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] : \text{type} \end{array} \quad ? : \begin{array}{c} \vdots \\ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \end{array} \quad \begin{array}{c} \vdots \\ \mathbf{wise}_s(\pi_1 u)[?/u] : \text{type} \end{array}}{\left[\begin{array}{c} u @ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \\ \mathbf{wise}_s(\pi_1 u) \end{array} \right] : \text{type}} \quad (@)$$

In order for the sentence (733) to satisfy SFC, the sentence must be uttered under a context Γ such that the following type inhabits a proof (namely, there exists a king of France).

$$(739) \quad \Gamma \vdash ? : \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right]$$

Recall the bridging hypothesis of DTS about empirical inferences: for any two sentences S_1 and S_2 , semantic representations of which are SR_1 and SR_2 , respectively, $S_1 \implies S_2$ holds if and only if, for any context Γ , if $\Gamma \vdash SR_1 \text{ true}$, then $\Gamma \vdash SR_2 \text{ true}$.

Therefore, an inference from (733) to (736) proceeds as follows: For (733) to be true, it must satisfy the SFC. But the SFC for (733) demands that the (736) be true under the context Γ . Consequently, if (733) is true, (736) must also be true in that same context Γ .

27.3.1 Projection out of Negation

Now, consider how the presupposition *France has a king* in (740) is projected out of negation.

(740) It is not the case that the king of France is wise.

The syntactic structure and semantic composition of (740) are given as follows.

(741) Syntactic structure of (740)

$$\frac{\frac{\text{It is not the case that}}{(S/S)^*} \quad \frac{\text{the king of France is wise}}{S^*}}{\frac{S^*}{S} \checkmark} >^*$$

(742) Semantic composition of (740)

$$\begin{array}{c}
 \frac{\text{the king of France is wise}}{\left[\begin{array}{c} u@ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \\ \mathbf{wise}_s(\pi_1 u) \end{array} \right]} \\
 \frac{\neg id \quad \left[\begin{array}{c} u@ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \\ \mathbf{wise}_s(\pi_1 u) \end{array} \right]}{\neg \left[\begin{array}{c} u@ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \\ \mathbf{wise}_s(\pi_1 u) \end{array} \right]} >^* \\
 \frac{\neg \left[\begin{array}{c} u@ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \\ \mathbf{wise}_s(\pi_1 u) \end{array} \right]}{\neg \left[\begin{array}{c} u@ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \\ \mathbf{wise}_s(\pi_1 u) \end{array} \right]} \checkmark
 \end{array}$$

To check the semantic felicity condition of (740), one is required to complete a proof construction as follows.*²

(743) Type check diagram of (740)

$$\begin{array}{c}
 \frac{\begin{array}{c} \vdots \\ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] : \text{type} \end{array} \quad ? : \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \quad \begin{array}{c} \vdots \\ \mathbf{wise}_s(\pi_1 u)[?/u] : \text{type} \end{array}}{\left[\begin{array}{c} u@ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \\ \mathbf{wise}_s(\pi_1 u) \end{array} \right] : \text{type}} \quad (\textcircled{a}) \\
 \frac{\left[\begin{array}{c} u@ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \\ \mathbf{wise}_s(\pi_1 u) \end{array} \right] : \text{type}}{\neg \left[\begin{array}{c} u@ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \\ \mathbf{wise}_s(\pi_1 u) \end{array} \right] : \text{type}} \quad (\perp F) \\
 \frac{\neg \left[\begin{array}{c} u@ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \\ \mathbf{wise}_s(\pi_1 u) \end{array} \right] : \text{type}}{\neg \left[\begin{array}{c} u@ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \\ \mathbf{wise}_s(\pi_1 u) \end{array} \right] : \text{type}} \quad (\Pi F)
 \end{array}$$

A comparison of (743) and (738) reveals that the what SFC requires for the former is identical to the latter: For the sentence (740) to be felicitous, its utterance context Γ must render (736) true. Let us confirm how this fact provides an account for the following inference.

(744) It is not the case that the king of France is wise.

$\Rightarrow$ There exists a king of France.

Following the discussion in the previous section, for (740) to satisfy SFC, Γ must satisfy (739) again. Since it is a necessary condition for (740) to be true, Γ satisfies (739). Therefore, (736) is true.

27.3.2 Projection out of Implication

The same process is triggered for the antecedent of a conditional sentence like (745).

(745) If the king of France is wise, people will be happy.

The syntactic structure and semantic composition for (745) are calculated as below.

*² Recall that negation is defined as an implication of the form $\neg A \equiv A \rightarrow \perp$, where $\perp$ is the bottom type that has no inhabitants. According to the (ΠF) -rule, the proposition A and its negation $\neg A$ have the same presupposition.

(746) Syntactic structure of (745)

$$\frac{\frac{\text{If}}{(S/S)^*/S} \quad \frac{\frac{\text{the king of France is wise}}{S^*} \checkmark}{S} >}{(S/S)^*} \quad \frac{\text{people will be happy}}{S^*} >^*}{\frac{S^*}{S} \checkmark} >^*$$

(747) Semantic composition of (745)

$$\frac{\frac{\text{If}}{\lambda p. (u : p) \rightarrow \boxed{id}} \quad \frac{\frac{\text{the king of France is wise}}{\left[\begin{array}{c} v@ \left[\begin{array}{c} y : e \\ \mathbf{KingOf}(y, fr) \end{array} \right] \\ \mathbf{wise}_{s_1}(\pi_1 v) \end{array} \right]}} \checkmark}{\left[\begin{array}{c} v@ \left[\begin{array}{c} y : e \\ \mathbf{KingOf}(y, fr) \end{array} \right] \\ \mathbf{wise}_{s_1}(\pi_1 v) \end{array} \right]} >}{\left(u : \left[\begin{array}{c} v@ \left[\begin{array}{c} y : e \\ \mathbf{KingOf}(y, fr) \end{array} \right] \\ \mathbf{wise}_{s_1}(\pi_1 v) \end{array} \right] \right) \rightarrow \boxed{id}} \quad \frac{\text{people will be happy}}{\boxed{\mathbf{happy}_{s_2}(\mathit{people})}}}{\left(u : \left[\begin{array}{c} v@ \left[\begin{array}{c} y : e \\ \mathbf{KingOf}(y, fr) \end{array} \right] \\ \mathbf{wise}_{s_1}(\pi_1 v) \end{array} \right] \right) \rightarrow \boxed{\mathbf{happy}_{s_2}(\mathit{people})}} >^*}{\left(u : \left[\begin{array}{c} v@ \left[\begin{array}{c} y : e \\ \mathbf{KingOf}(y, fr) \end{array} \right] \\ \mathbf{wise}_{s_1}(\pi_1 v) \end{array} \right] \right) \rightarrow \boxed{\mathbf{happy}_{s_2}(\mathit{people})}} \checkmark}{\left(u : \left[\begin{array}{c} v@ \left[\begin{array}{c} y : e \\ \mathbf{KingOf}(y, fr) \end{array} \right] \\ \mathbf{wise}_{s_1}(\pi_1 v) \end{array} \right] \right) \rightarrow \mathbf{happy}_{s_2}(\mathit{people})} \checkmark}$$

(748) Type check diagram of (745)

$$\frac{\frac{\begin{array}{c} \vdots \\ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] : \text{type} \end{array} \quad ? : \frac{\begin{array}{c} \vdots \\ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \end{array} \quad \mathbf{wise}_{s_1}(\pi_1 v)[?/v] : \text{type}}{\left[\begin{array}{c} v@ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \\ \mathbf{wise}_{s_1}(\pi_1 v) \end{array} \right] : \text{type}} \quad \frac{\begin{array}{c} \vdots \\ \mathbf{happy}_{s_2}(\mathit{people}) : \text{type} \end{array}}{\left(u : \left[\begin{array}{c} v@ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \\ \mathbf{wise}_{s_1}(\pi_1 v) \end{array} \right] \right) \rightarrow \mathbf{happy}_{s_2}(\mathit{people}) : \text{type}} \quad \text{(\textcircled{a})}}{\left(u : \left[\begin{array}{c} v@ \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \\ \mathbf{wise}_{s_1}(\pi_1 v) \end{array} \right] \right) \rightarrow \mathbf{happy}_{s_2}(\mathit{people}) : \text{type}} \quad (\Pi F)$$

What is presupposed by (745) can be read off from the proof search required by the above type checking.

$$(749) \quad \Gamma \vdash \left[\begin{array}{c} x : e \\ \mathbf{kingOf}(x, fr) \end{array} \right] \text{ true}$$

Thus, (745) can be felicitously uttered only under a context Γ that satisfy the above, therefore we can conclude that the a king of France exists under Γ .

Proposition 465 (Presupposition Projection in DTS). A presupposition projects because it's truth is a requirement for a sentence containing it to be *semantically well-formed*, not to be *true*.

Corollary 466. Existence of an antecedent is a requirement for a sentence containing anaphora to be semantically well-formed, not to be true.

27.4 Presupposition Filtration

Consider a proposition B that contains a presupposition trigger, yielding a presupposition P . If another proposition A entails P , then it is a known property of constructions such as $A \times B$ and $A \rightarrow B$ that the aforementioned contextual requirement on Γ (i.e., $\Gamma \vdash P \text{ true}$) vanishes.^{*3} This is the phenomenon known as *presupposition filtration* (or *presupposition filtering*), which is illustrated by the following examples.

- (750) a. France has a king and **the king of France** is wise.
 b. If France had a king, **the king of France** is wise.

In both the conjunction and the conditional constructions, the overall proposition no longer requires that *There exists a king of France* be true within the context Γ . The present analysis by DTS can explain the filtering of presupposition without further stipulation. The syntactic structure and semantic composition of (750b) are derived as follows.

^{*3} Presupposition filtering poses a challenge for the formalization of semantic phenomena in the form of inferences. This difficulty arises because the cancellation of a presupposition is not a matter of cancellation of an inference. For instance, a sentence with presupposition and its conjunctive-filtering counterpart often share the same logical consequence. Consider the following:

- (1) a. The king of France is wise. $\implies$ There exists a king of France.
 b. France has a king, and the king of France is wise. $\implies$ There exists a king of France.

Both of these sentences trigger the presupposition that France has a king. The truth-conditions for both are identical, as they are true if and only if both *France has a king* and *The king of France is wise* are true.

The distinction between these sentences may be captured by examining their behavior within a context that does not entail the presupposition, but can be updated by it. In the following dialogues, sentence (2) is pragmatically infelicitous as a cooperative utterance – B uses a presuppositional sentence when A's ignorance of the presupposition is explicit – whereas sentences (3) and (4) are not.

- (2) A: I'm not sure if there exists a king of France.
 B: #The king of France is wise!
 (3) A: I'm not sure if there exists a king of France.
 B: France has a king, and the king of France is wise.
 (4) A: I'm not sure if there exists a king of France.
 B: If France has a king, then the king of France is wise.

While it is possible to formalize this distinction in terms of pragmatic felicity, it is important to note that sentence (2) can still be felicitous within a non-cooperative or confrontational context. This pragmatic nuance underscores the difficulty of modeling presupposition filtering as its inferential behaviour.

hypothesis, utilizing underspecified types to offer a unified account for both phenomena.

(754) Presupposition filtering

- a. John has children and **John's children** are wise.
- b. If John has children, **John's children** are wise.

Compare (754) and (??) with the paradigm examples of anaphora.

(755) Anaphora binding

- a. John owns a donkey and he beats **it**.
- b. If John owns a donkey, he beats **it**.

A key piece of evidence supporting the PIA hypothesis is the observation that presuppositions, much like anaphora, are free to select their antecedents (or filters). Consider the following sentence.

(756) If John has grandchildren, his children must be happy. ((27) in van der Sandt (1992), p.349)

In this sentence, *his children* acts as a presupposition trigger. If we assume that *his* refers to *John*, a presupposition is introduced: that John has a child. Since this presupposition is entailed by the antecedent clause *John has grandchildren*, it can potentially be filtered.

However, this sentence is in fact ambiguous; it permits both a reading where the presupposition is filtered and one where it is not. The former arises through the process of filtering, while the latter is a result of non-filtering. As van der Sandt (1992) points out, the non-presuppositional reading becomes salient when the sentence is followed by a continuation such as below.

(757) If John has a grandchildren, his children must be happy. They wanted to have offspring long ago. ((351) in van der Sandt (1992), p.351)

As this example illustrates, the application of filtering is optional, not obligatory. This observation parallels the fact that an potential antecedent for an anaphoric expression is not strictly required to do so.

In DTS, this distinction between the two readings is modeled as a difference in proof constructions. The syntactic structure, semantic representation, and type checking for the expression in question are as follows:

(ToDo: syntactic structure)
(ToDo: semantic composition)

Consequently, for the SFC to hold, the context Γ must satisfy the following relation:

(758) $\Gamma, \text{John has a grandchildren} \vdash \text{John has a children}$

The assumption of world knowledge regarding grandchildren – namely,

(759) $(x : e) \rightarrow (z : e) \rightarrow \mathbf{grandChildOf}(x, z) \rightarrow (y : e) \times \mathbf{childOf}(x, y) \times \mathbf{childOf}(y, z)$

is not ad-hoc, as it constitutes the very definition of the concept of grandchildren. Therefore, $\mathbf{grandChildOf}(\text{john}, x)$ implies:

(760) $((y : e) \times \mathbf{childOf}(\mathit{john}, y))$

It is crucial to note, however, that the SFC does not specify the precise method for proving that John has children. That is, this conclusion could be derived without relying on Γ by leveraging the above-stated relation, or alternatively, one could stipulate that $\Gamma \vdash$ John has children holds independently. The latter approach would entail establishing the existence of a child distinct from the parent of the grandchildren.*4

This example illustrates how presuppositions are projected in DTS, specifically delineating what the SFC imposes (and does not impose) on a given context. The core argument is that a sentence’s presupposition is the proposition found to the right of the “@” symbol within an underspecified type. The original context, Γ , may be augmented with information from other clauses – such as the first conjunct or the antecedent of a conditional – which can relax the constraints on Γ . Crucially, however, this information is not considered part of the presupposition itself.

27.6 The Proviso Problem

The proviso problem (Geurts, 1996, 1999; Schlenker, 2011) is a significant issue in the analysis of presuppositions, particularly for theories that employ a filtering or satisfaction-based approach. We can illustrate the problem by examining the contrasting behaviors of two sentences with similar logical structures.

File Change Semantics (FCS), as developed by Heim (1983a), provides a classic example of this issue. In FCS, presuppositions are not merely checked against the context; they are assumed to be compositionally integrated into the meaning of the sentence. For a conditional statement of the form $A \rightarrow B$, where B carries a presupposition P , FCS predicts that the presupposition of the entire conditional is $A \rightarrow P$. This mechanism seems to work for certain cases. Consider the sentence:

(761) If John is a scuba diver, he will bring his wetsuit.

The term *his wetsuit* presupposes *John has a wetsuit*. According to FCS, this is filtered by the antecedent *John is a scuba diver*, and the resulting presupposition of the whole sentence is:

(762) If John is a scuba diver, then he has a wetsuit.

Given our world knowledge that scuba divers typically own wetsuits, this conditional presupposition is uninformative and holds true, which correctly explains why the original sentence appears to have no presuppositions. However, a problem arises with an almost identical sentence structure:

(763) If John flies to London, his sister will pick him up.

Here, the term *his sister* triggers the presupposition *John has a sister*. The FCS analysis predicts the presupposition of the entire sentence to be:

*4 In principle, there could exist a derivation path where the premise Γ combined with *John has grandchildren* would license the inference *John has children*, though this specific example does not illustrate the point saliently.

However, one may consider an alternative scenario: a country has legislated that citizens may not have a child unless they have either siblings or they are in a familial relation to the royal family. In such a context, if Γ proves that John is not related to the royal family, we can then infer that John has children.

(764) If John flies to London, then he has a sister.

This result is counterintuitive. Intuitively, the sentence *If John flies to London, his sister will pick him up* does not seem to carry the complex presupposition that a relationship exists between *flying to London* and *having a sister*. The core of the proviso problem is to explain this difference.

In contrast, the DTS approach offers a straightforward solution (Yana et al., 2021). From the DTS perspective, the presuppositions of the two sentences are simply:

- (765) a. John has a wetsuit.
b. John has a sister.

The satisfaction condition for these presuppositions is given by the following judgments:

- (766) a. $\Gamma, \text{John is a scuba diver} \vdash \text{John has a wetsuit}$
b. $\Gamma, \text{John flies to London} \vdash \text{John has a sister}$

Notice that, up to this point, the analyses of the two sentences are identical. Both require that the presupposition can be derived from the context Γ plus the information from the antecedent of the conditional.

Crucially, the DTS framework does not care about the path of the proof. The question of whether or not the proof of *John has a wetsuit* utilizes the additional information (*John is a scuba diver*) is a matter of proof search, not a part of the sentence's meaning. The satisfaction condition is met as long as a valid proof exists, with or without the antecedent information.

In essence, DTS maintains that the meaning of the sentence specifies a relationship that must hold between the presupposition and the context, but it does not dictate how that relationship should be established given our world knowledge. This analysis provides a unified and elegant explanation for both cases and suggests that the proviso problem is a pseudo-problem, a byproduct of an inadequate formalization of presupposition.

27.7 Global and Local Accommodations

Consider (767), taken from Heim (1983b).^{*5}

(767) No nation cherishes its king.

The sentence (767) is interpreted in two distinct ways regarding its presupposition, as described in Heim (1983b, p. 403) as follows:

What about quantifiers other than universal? Concerning “no,” we find conflicting factual claims in the literature. According to Cooper (1983), (24) [=our (767)] should presuppose that every nation (in the relevant domain of discourse) has a king; for ?, it presuppose that some nation does.

One reading of (767) that implies **every nation has a king** is obtained via the *global accommodation*, whereas another reading of (767) implying **Some nation has a king** is obtained via the *local accommodation*.

^{*5} Sentence (24) on p. 403.

We assume that the lexical item of **its** is defined as (??) in almost the same way as **his** in (??) except for its presuppositional content:

Since the syntactic structure of (767) is almost the same as (555a), we obtain its semantic representation via the homomorphic semantic composition, which is (768) corresponding to the BVA reading for (767).

$$(768) \quad \left(w : \begin{bmatrix} x : e \\ \mathbf{nation}(x) \end{bmatrix} \right) \rightarrow \neg \begin{bmatrix} u@ \begin{bmatrix} x : e \\ \neg \mathbf{human}(x) \end{bmatrix} \\ v@ \begin{bmatrix} y : e \\ \mathbf{king}(y) \times \mathbf{of}(y, \pi_1 u) \\ \mathbf{cherish}(\pi_1 v, \pi_1 w) \end{bmatrix} \end{bmatrix}$$

The type checking for (768) factors through (769).

$$(769) \quad \Gamma, w : (x : e) \times \mathbf{nation}(x) \vdash \begin{bmatrix} u@ \begin{bmatrix} x : e \\ \neg \mathbf{human}(x) \end{bmatrix} \\ v@ \begin{bmatrix} y : e \\ \mathbf{king}(y) \times \mathbf{of}(y, \pi_1 u) \\ \mathbf{cherish}(\pi_1 v, \pi_1 w) \end{bmatrix} \end{bmatrix} : \text{type}$$

which will invoke the @-rule (??) and call for the type checking (770a), which obviously succeeds, and then the proof search (770b).

$$(770) \quad \begin{array}{ll} \text{a. } \Gamma, w : \begin{bmatrix} x : e \\ \mathbf{nation}(x) \end{bmatrix} \vdash \begin{bmatrix} x : e \\ \neg \mathbf{human}(x) \end{bmatrix} : \text{type} \\ \text{b. } \Gamma, w : \begin{bmatrix} x : e \\ \mathbf{nation}(x) \end{bmatrix} \vdash ? : \begin{bmatrix} x : e \\ \neg \mathbf{human}(x) \end{bmatrix} \end{array}$$

If we assume world knowledge that every nation is non-human (i.e., the constant term **nnh**: $(x : e) \rightarrow \mathbf{nation}(x) \rightarrow \neg \mathbf{human}(x)$), the proof search for (770b) will find the term $(\pi_1 w, \mathbf{nnh}(\pi_1 w)(\pi_2 w))$ that corresponds to the BVA reading of (767). The next step is the type checking (771), which breaks down to the type checking (772a) and the proof search (772b).

$$(771) \quad \Gamma, w : \begin{bmatrix} x : e \\ \mathbf{nation}(x) \end{bmatrix} \vdash \begin{bmatrix} v@ \begin{bmatrix} y : e \\ \mathbf{king}(y) \times \mathbf{of}(y, \pi_1 w) \\ \mathbf{cherish}(\pi_1 v, \pi_1 w) \end{bmatrix} \end{bmatrix} : \text{type}$$

$$(772) \quad \begin{array}{ll} \text{a. } \Gamma, w : \begin{bmatrix} x : e \\ \mathbf{nation}(x) \end{bmatrix} \vdash \begin{bmatrix} y : e \\ \mathbf{king}(y) \times \mathbf{of}(y, \pi_1 w) \end{bmatrix} : \text{type} \\ \text{b. } \Gamma, w : \begin{bmatrix} x : e \\ \mathbf{nation}(x) \end{bmatrix} \vdash ? : \begin{bmatrix} y : e \\ \mathbf{king}(y) \times \mathbf{of}(y, \pi_1 w) \end{bmatrix} \end{array}$$

Again, (772a) is a routine, and the proof search for (772b) is a search for the king of y . Remember that, in the case of (565) and (570b), we assumed that there exists a world knowledge **fatherOf**, that is, every boy has a father. The question here is what happens if we do not have a knowledge that every nation has a king, and thus the proof searching for (772b) fails.

The first strategy for our type checker is global accommodation, defined in Bekki (2014), which can be restated as the following instruction.

Definition 467 (Global accommodation). When the proof search

$$\llbracket \Gamma \vdash ? : A \rrbracket$$

fails for the type A such that $fv(A) = x_1, \dots, x_n$, the type checker may add the constant term of type $(x_1 : A_1) \rightarrow \dots \rightarrow (x_n : A_n) \rightarrow A$ (where $x_1 : A_1, \dots, x_n : A_n \subseteq \Gamma$) to the signature and re-run the type checking.

Applying Definition 467 to the case of (772b), the constant term added to the signature is $\left(w : \left[\begin{array}{l} x : e \\ \mathbf{nation}(x) \end{array} \right] \right) \rightarrow \left[\begin{array}{l} y : e \\ \mathbf{king}(y) \times \mathbf{of}(y, \pi_1 w) \end{array} \right]$, which is exactly the knowledge that every nation has a king.

The second strategy for our type checker is local accommodation, which, under the definition of underspecified types, is given the following definition.

Definition 468 (Local accommodation). When the proof search

$$\llbracket \Gamma \vdash ? : A' \rrbracket$$

required for the type checking $\llbracket \Gamma \vdash (x@A) \times B : \mathbf{type} \rrbracket$ fails, the type checker may replace the result of $\llbracket \Gamma \vdash (x@A) \times B : \mathbf{type} \rrbracket$ with the result of type checking $\llbracket \Gamma \vdash (x : A) \times B : \mathbf{type} \rrbracket$.^{*6}

One benefit of employing the underspecified types is that it allows us to define the operation of local accommodation in a straightforward way. This is due to their definition that takes a closer form to that of the Σ -formation rule: when the proof search $\llbracket \Gamma \vdash ? : A \rrbracket$ fails in the type checking $\llbracket \Gamma \vdash (x@A) \times B : \mathbf{type} \rrbracket$, the semantic system may optionally replace $(x@A) \times B$ with $(x : A) \times B$ and re-run the type checking. Note that this transformation is safe owing to the similarity between the verification conditions of the @-rule and Σ -formation rule.

In other words, the local accommodation in DTS is understood as accommodating the presupposed content of an underspecified type by existentially quantifying it. Applying Definition 468 to (772b), replacing the type checking for (771) with the following:

$$(773) \quad \Gamma, w : \left[\begin{array}{l} x : e \\ \mathbf{nation}(x) \end{array} \right] \vdash \left[\begin{array}{l} v : \left[\begin{array}{l} y : e \\ \mathbf{king}(y) \times \mathbf{of}(y, \pi_1 w) \end{array} \right] \\ \mathbf{cherish}(\pi_1 v, \pi_1 w) \end{array} \right] : \mathbf{type}$$

thus the whole semantic representation for (767) turns into (774).

^{*6} This operation should not be applied to all types of anaphora and presupposition: for example, it is known that pronouns in general do not undergo local accommodation when their antecedents are missing. For this purpose, we may want to add a binary feature to underspecified types that tells us whether local accommodation is applicable to them, an approach pursued by Yana et al. (2024). Alternatively, we may argue that the prohibition of local accommodation for pronouns is based rather on pragmatic factors. This is a controversial issue and I would like to leave it open.

$$(774) \quad \Gamma \vdash \left(w : \left[\begin{array}{l} x : e \\ \mathbf{nation}(x) \end{array} \right] \right) \rightarrow \neg \left[\begin{array}{l} v : \left[\begin{array}{l} y : e \\ \mathbf{king}(y) \times \mathbf{of}(y, \pi_1 w) \end{array} \right] \\ \mathbf{cherish}(\pi_1 v, \pi_1 w) \end{array} \right]$$

In words, (774) claims that No **nation** has a **king** and **cherishes** it, which exactly corresponds to the locally accommodated reading of (767).

27.8 Presupposition Cancellation and Meta-linguistic Negation

In a certain case, presuppositions can be *cancelled* within the scope of a negation, a phenomenon known as *presupposition cancellation*. Consider the classic example:

(775) The king of France is NOT wise, since there is no such thing as “the king of France.”

In this case, the presupposition of a French king’s existence is not required by the context. However, two important observations must be made on this:

1. This type of negation, often marked by a specific intonation, is termed *metalinguistic negation*.
2. Presupposition cancellation of this kind does not occur in affirmative sentences.

The following example illustrates the second point.*⁷

(776) The king of France is wise, #but actually, there is no such thing as “the king of France.”

In DTS, metalinguistic negation is analyzed as a signal that local accommodation has occurred within the scope of the negation. In the example above, local accommodation would modify SR to one where the presupposition is integrated directly into the negated clause. This local integration ensures that the existence of a French king is not projected out of the negation. The theory also correctly predicts that local accommodation would not manifest in affirmative sentences, as this behavior is unique to the negation operator.

History and Further Readings

- Early works: Frege (1892) Russell (1905) Strawson (1950)
- Presupposition projections and filtration: Stalnaker (1973, 1974) Gazdar (1979a,b) Heim (1983b)

*⁷ The second point refutes the notion that presupposition is merely a tendency. Introducing a mechanism for local accommodation into a semantic theory inevitably compromises its testability. When predicting whether a given sentence presupposes a certain proposition, the ever-present option of local accommodation means the proposition may or may not hold in the given context, which can render the prediction unfalsifiable. However, we can still maintain a degree of empirical accountability. This is because presuppositions do not disappear in affirmative sentences, metalinguistic negation is often distinguished by intonation, and certain negative morphemes are resistant to a metalinguistic reading. Furthermore, the existence of metalinguistic negation does not affect the projection properties of conditional sentences.

- Presupposition is anaphora hypothesis: van der Sandt and Geurts (1991) van der Sandt (1992)
- Type-theoretic approaches: Ranta (1994) Mineshima (2008) Bekki and Mineshima (2017)